ModelIt!

Student Activity Pack

Can Cities Run on 100% Clean Energy?

Modeling Renewable Grid Stability, Storage, and Intermittency

Grade 11L2 | NGSS HS-PS3-3, HS-ESS3-2 | Systems and System Models / Energy and Matter

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **What do you think happens to a city's electricity when the sun goes down if the city relies on solar power?**

2. **Why can't we just build enough solar panels and wind turbines to power everything?**

3. **Draw a graph showing when you think a city uses the most and least electricity during a 24-hour day.**

4. **What technologies do you know about that can store large amounts of energy?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Solar Generation Capacity | Wind Generation Capacity | Energy Storage Capacity | Grid Demand Pattern | Supply-Demand Balance | Grid Reliability Index

EXTERNAL	INTERNAL

? Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: 50% Renewable

Half renewable, half conventional, moderate storage

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: 90% Renewable

Heavy renewable, minimal fossil backup, current storage

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: 100% Renewable + Storage

All renewable with 48 hours of storage

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Demand Response Programs — Smart grid systems that incentivize consumers to shift electricity usage away from peak hours — like running dishwashers at midnight when wind power is abundant — reducing the gap between supply and demand

Green Hydrogen Production — Using excess renewable electricity to split water into hydrogen and oxygen through electrolysis — storing energy in chemical form for weeks or months and converting it back to electricity when needed

Geographic Diversification — Connecting grids across wide geographic areas so that when wind is calm in one region, power can be imported from where it's blowing — the wind is always blowing somewhere

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If the sun doesn't always shine and the wind doesn't always blow, can an entire city really run on 100% renewable energy without the lights going out? Use evidence from your simulation to support your answers.

1 Why does the difficulty of integrating renewables increase exponentially rather than linearly as you approach 100%?

2 How does the duck curve in your model create both waste and shortage on the same day?

3 What does your model reveal about why seasonal storage is a fundamentally different challenge than daily storage?

4 How would climate change itself (hotter summers, more extreme weather) affect the grid your model designs?

5 What trade-offs does your model expose between cost, land use, reliability, and speed of decarbonization?

8 STEM Challenge: Design a 100% Renewable City Energy Plan

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YOUR ENGINEERING MISSION

Design a comprehensive energy plan for a mid-sized city (population 500,000) that achieves 100% renewable electricity while maintaining 99.97% grid reliability through all weather conditions.

THE SCENARIO

A city council has committed to 100% renewable electricity by 2040 and hired your engineering team to design the generation and storage portfolio. You must balance solar, wind, and storage to maintain reliability through the worst weather weeks of the year while minimizing cost and land use.

Engineering Design Thinking

1. WHAT IS THE OPTIMAL RATIO OF SOLAR TO WIND CAPACITY FOR YOUR CITY'S LOCATION AND CLIMATE?
2. HOW MANY HOURS OF BATTERY STORAGE WOULD YOU DEPLOY VERSUS LONG-DURATION STORAGE LIKE PUMPED HYDRO?
3. WHAT DEMAND FLEXIBILITY PROGRAMS WOULD YOU IMPLEMENT TO SHIFT LOADS AWAY FROM EVENING PEAKS?
4. HOW WOULD YOU PHASE THE TRANSITION OVER 15 YEARS TO MAINTAIN RELIABILITY AT EACH STAGE?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Grid Integration Engineers design the systems that connect renewable energy sources to electrical grids while maintaining stability and reliability. They work for utilities, grid operators like CAISO and ERCOT, renewable energy companies, and consulting firms, earning $85,000–$155,000/year. Energy Storage Engineers specializing in battery systems and grid-scale storage earn $90,000–$160,000/year.